



## Problems (Black Scholes Option Pricing)

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For all the problems below, assume Black Scholes world with no dividends and European options. The call option is denoted as  $C$ , and the put option is denoted as  $P$ . All other symbols have their usual meanings.

Prove the following using no-arbitrage arguments. Do not use Black-Scholes equation.

1. (Put-Call Parity)  $P_0 + S_0 = C_0 + Ke^{-rT}$

2. (Bounds) Show that call and put options have the following bounds

$$\max(0, S_0 - Ke^{-rT}) \leq C_0 \leq S_0$$

$$\max(0, Ke^{-rT} - S_0) \leq P_0 \leq Ke^{-rT}$$

3. (Monotonicity in Maturity) Show that both call and put option prices are higher with higher maturities, everything else considered equal. In other words, if  $T_2 > T_1$

$$C_0(S_0, K, T_1) < C_0(S_0, K, T_2)$$

$$P_0(S_0, K, T_1) < P_0(S_0, K, T_2)$$

4. (Monotonicity in Strike) If  $K_2 > K_1$ , assuming all else is equal, show that

$$C_0(S_0, K_1, T) > C_0(S_0, K_2, T)$$

$$P_0(S_0, K_2, T) > P_0(S_0, K_1, T)$$

5. If  $K_2 > K_1$ , Show that  $C_0(S_0, K_1, T) - C_0(S_0, K_2, T) \leq (K_2 - K_1)e^{-rT}$

For the following questions you can use Black-Scholes equation.

6. (Infinitesimal maturity) Show that

$$\lim_{T \rightarrow 0} C_0 = \max\{S_0 - K, 0\}$$

$$\lim_{T \rightarrow 0} P_0 = \max\{K - S_0, 0\}$$

7. (Infinitesimal volatility) Show that

$$\lim_{\sigma \rightarrow 0} C_0 = \max\{S_0 - Ke^{-rT}, 0\}$$

$$\lim_{\sigma \rightarrow 0} P_0 = \max\{Ke^{-rT} - S_0, 0\}$$

8. (Infinite volatility) Show that

$$\lim_{\sigma \rightarrow \infty} C_0 = S_0$$

$$\lim_{\sigma \rightarrow \infty} P_0 = Ke^{-rT}$$

9. (Invariance I) What is the ratio of option prices in the following two cases

- a. Case I - Double the Stock price
- b. Case II - Halve the Strike price

10. (Invariance II) What happens to the option price if both Stock price and Strike price are doubled?

11. (Put-Call Supersymmetry) An analyst wants to calculate the price of a Call option. He can see that the price of the Put counterpart is \$10. But the analyst immediately notices that due to a technical glitch, the stock price, strike price and the volatility numbers have been entered with a negative sign in the Put valuation. What is the correct price of the call option?

12. What is the price of a European option with payoff

$$\max\{S_T^2 - K, 0\}$$